
BUSINESS INTELLIGENCE
SEASON 10
ROUND 2 : SYNTHESIZE
TEAM: SHOOTING STAR

EXECUTIVE SUMMARY

we found that...

S **SITUATION**

1,000,528 customers (99.98%) applied in the Dec-24/Jan-25 Tết wave (12× a normal month). The company has five data sources but no single view of who can carry credit, who wants to come back, and what to do with each customer.

C **COMPLICATION**

Money and attention go to the wrong places: 37% of outbound dials reach 113,644 customers who have never answered; one lead source (RA) produces 54% of leads but 0.8% of deals; first-time Tết borrowers are 2.1× as risky as repeat borrowers. Meanwhile 58% of "inactive" customers are not lost (87.6%) are one-time Tết borrowers.

Q **KEY QUESTION**

Whom should we grow, whom should we protect, and where can we cut cost. Is it customer by customer?

A **ANSWER**

Two explainable 0–100 scores (financial health, engagement) route every customer to one of 11 actions. The health score ranks a period it never saw as well as a black-box model, the lowest-score decile is 8.9× riskier than the highest

8.9×
risk gap, lowest vs highest score decile (tested on the Tết wave)

≈2.8 bn VND/yr
saved by stopping dials after 5 unanswered calls (base case*; 1.2–4.9 bn range)

54% → 0.8%
RA share of leads vs deals: re-target lead budget

392,164
one-time Tết borrowers: one reminder in early December, not a campaign now

60,597
customers with real repayment stress → support queue (45,053 by SMS/app, no calls)

*assumption-based, range 1.2–4.9 bn

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No unified view of 1,000,528 customers: three challenges, one action per customer

The business problem

Sells to the wrong people

credit offers reach customers who cannot carry them

Forgets the right people

healthy customers go quiet with no reason to return

Wastes contact budget

calls and leads go where they do not convert

Table 1: Five things the company does not know

Unknown	Business question	Answered by
Health	How financially healthy is each customer?	C1: Health Score
Engagement	How engaged is each customer?	C2: Engagement Score
Potential	Who has high future potential?	C3: Matrix
Stress	Who may be financially stressed?	C1 + C3
Action	What should we do for each customer?	C3: Next Best Action

Insight

The file is one cohort: **99.98%** of customers applied in the Dec-24/Jan-25 Tết wave, so "quiet" often means seasonal, not lost.

Table 2: What the brief asks and what we deliver

Challenge	Requirement from the brief	Our solution
C1: Financial Health	Factors, a 0 – 100 score, segments, +/- drivers, justification	5-pillar points scorecard on a risk-anchored scale, tested out of time
C2: Engagement	Recency, frequency, interaction, leads, products; link to outcomes	5 transparent components, validated on later applications
C3: Segments & NBA	Combine scores, find opportunities, recommend and explain an action	3 × 3 matrix plus contact guards; one action and reason per customer

Our approach

AUDIT
5 sources, data traps fixed

→

C1 HEALTH
who can carry a loan

→

C2 ENGAGE
who will come back

→

C3 ACTION
what to do with each one

1,000,528
rows in one CSV: scores, segment, action, reason

5 scripts
reproducible; 13 unit tests and 7 QA checks pass

1 dashboard
to explore every score and action

Three data traps would have broken every score: duplicates, censoring and scrambled call dates

Table 3: Five sources at a glance

Source	Rows	Cover	Window
CONTRACT	6,234,093	100%	04/09–06/25
CUSTOMER	1,000,528	100%	static
LEADS	2,143,891	64.5%	10/19–06/25
PAST_BEHAVIOR	4,412,610	47.3%	12/22–06/25
CALL	4,000,000	36.1%	01/25–08/26

Figure 1: Loan applications per month, Jan 2023 to Jun 2025

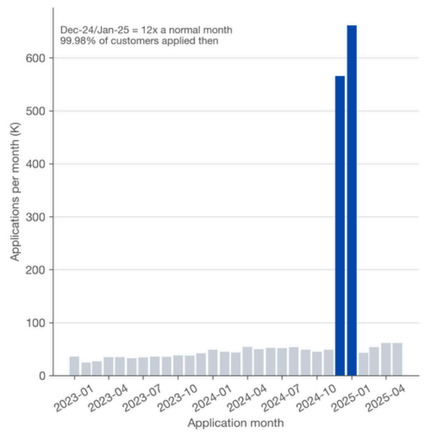


Figure 2: Ratio of 4th - 3rd instalment late rates by signing week

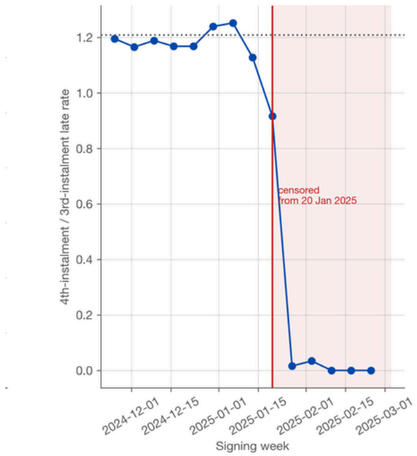


Table 4: Data traps, evidence and treatment

Trap evidence	Treatment
Duplicated contracts: 1,953,716 rows (31%) are copies carrying income = 0	Dedupe to 4,280,377 cases; otherwise counts inflate 1.46×
Right-censored labels: 4th/3rd late ratio falls from 1.21 to ~0 for loans signed from 20 Jan 2025	Every label and contract comparison stops at 20 Jan 2025
Call dates scrambled, all call types: only 4.1% of "sign successful" calls fall within 14 days of a real signing; 2.2% of repeat calls are under 1 day apart	Never use call dates; use per-customer call counts and outcomes only
One cohort, not the book: 99.98% applied in Dec-24/Jan-25, 32% for the first time	Validate on the Tết wave, "applied after Dec-24" is never an outcome
Unusable fields: FLAG_CAR_OWNER 100% "X"; household income 0.9% filled; dependents 98.5% empty; incomes up to 9bn VND	Excluded; income capped at p99.9; personal income is the DTI base
"Paid off" is not good: 82% had a late flag (Finished: 1.4%)	Treated as stressed history

Leakage guards (no future information in any score)

Write-offs count only on loans ≥ 12 months old, late flags only after 120 days, open loans set by maturity date, not snapshot status · **MAX_OFFER** is the lender's own decision, so its weight is capped at 13% to avoid a self-confirming loop.

Five explainable pillars decide the score, and every factor's direction is visible

Table 5: Pillars, weights and drivers (- lowers) / (+ raises) the score

Pillar	Weight	Key factors and observed stress (training windows)
Repayment History	35%	- Past late loans: 1.01% (none) → 20.3% (3+ or write-off), late loan < 6 months ago: 18.2%, + 8+ clean loans: 0.43%
Credit Standing	26%	- 60%+ of applications rejected: 2.70% vs 1.02%, + 8+ years of history: 1.12% vs 1.56%
Debt Burden (DTI)	13%	- Instalments above 22.5% of personal income: stress climbs 4.5× faster (Figure 3)
Down-payment Buffer	13%	+ Larger down payment: TW loans with 10 - 30% down are 2.3× as stressed as 50%+
Pre-approved Capacity	13%	+ Lender's largest pre-approved offer: 0.87% at 80M+ VND vs 3.32% under 10M

Each point has a meaning: the 0 - 100 scale is anchored to risk

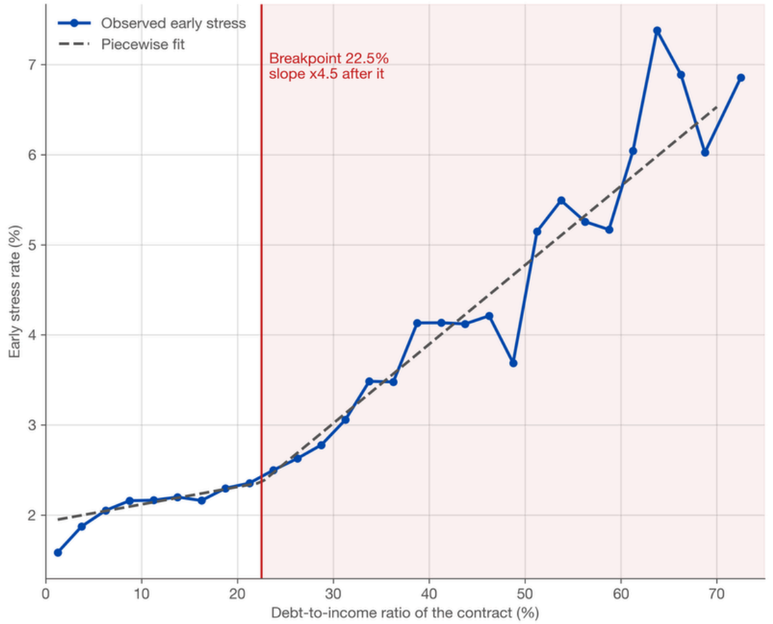
80 pts
= 0.5× average stress (1.57%)

50 pts
= 1.5× average stress (4.72%)

+18 pts
halves the odds of early stress

16.3%
stress of the 0.2% hit by policy caps

Figure 3: Early stress vs debt-to-income: flat until DTI ≈ 22.5%, then +0.88% per 10 points (vs +0.19%)



Kept out of the score, on purpose

- **Insurance:** insured loans are more stressed in all 12 product × DTI cells penalising it would discourage protection.
- **Open-loan count:** 3+ open loans look safer (1.99% vs 2.43%) only because the lender tops up clients it trusts, leverage enters via DTI.

CHALLENGE 1 - CUSTOMER FINANCIAL HEALTH SCORE

The score ranks a period it never saw: the worst decile is 8.9× as stressed as the best

***Figure 4*: Observed stress by score decile (bars) vs stress expected from the score (red)**

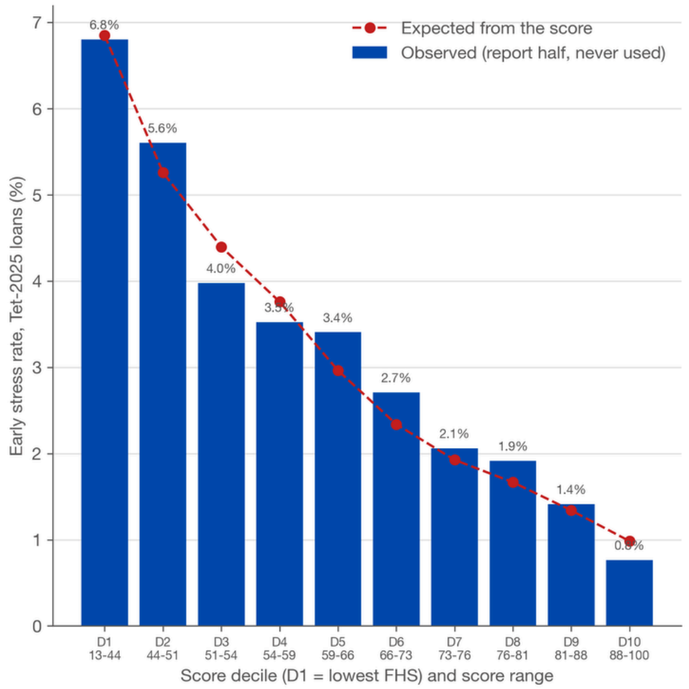
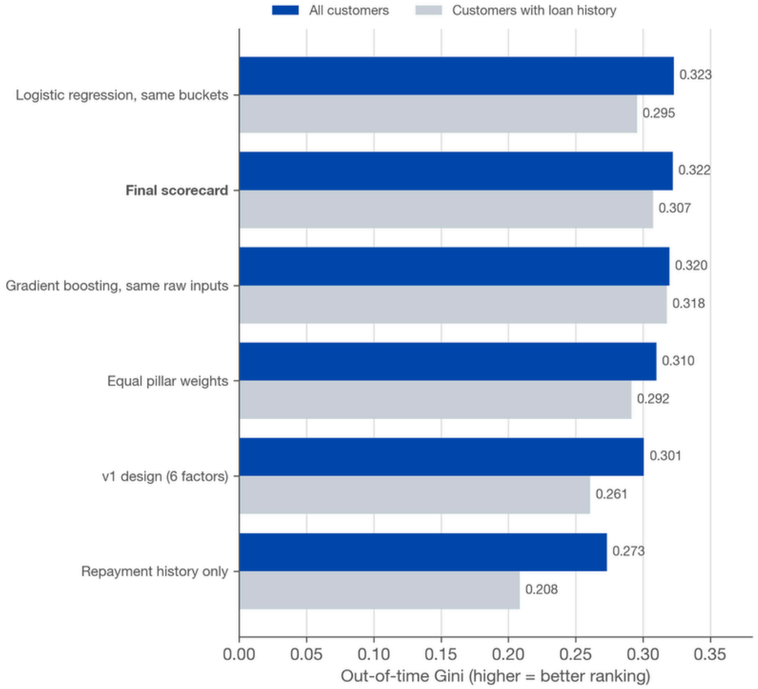


***Figure 5*: Out-of-time Gini, final scorecard vs simpler and more complex challengers**



Gini 0.322
KS 0.238 - AUC 0.661 on the untouched report half

0.301 - 0.310
Gini with any one pillar dropped: no single pillar carries it

≥ 0.987
rank correlation under 200 random ±25% weight shifts (Gini 0.313 - 0.323)

PSI 0.064
train → test; 0.104 test → today: a stable population

METHOD RATIONALE

Simple is as accurate: Logistic regression re-estimating every weight gets 0.323 and gradient boosting 0.320: complexity buys nothing here.

Calibrated: Expected stress per decile tracks the observed bars, so the 0 - 100 number doubles as an expected stress rate.

Better than our v1: Gini 0.322 vs 0.301 (with history: 0.307 vs 0.261) after the censoring fix and the thin-file split.

CHALLENGE 1 - CUSTOMER FINANCIAL HEALTH SCORE

We sort customers into three groups by how likely they are to fall behind on payments, and check that "risky" is not just "new".

Figure 6: **Financial Health Score today, 1,000,528 customers by tier**

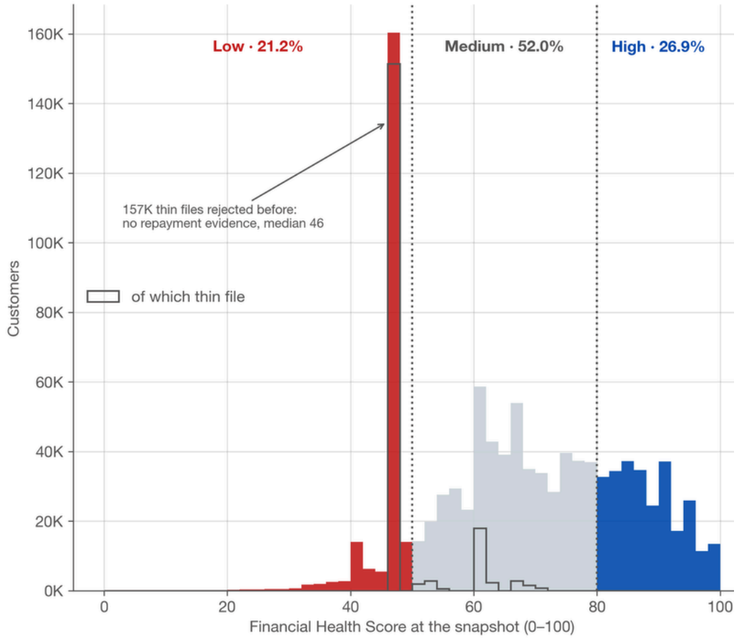


Figure 7: **Out-of-time stress by band: a 7.5× spread**

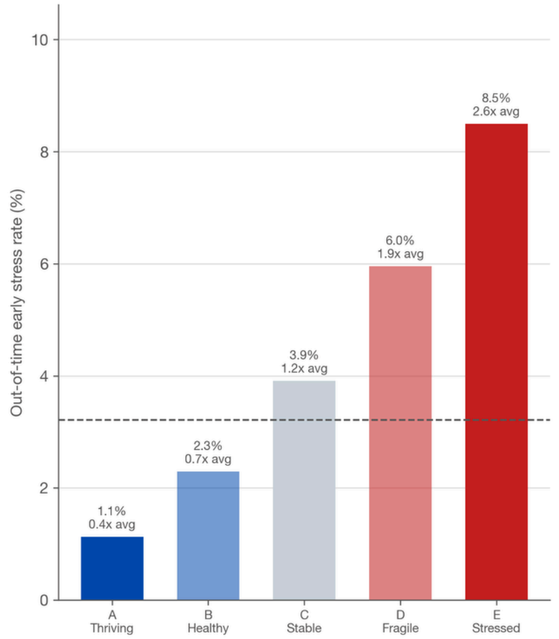
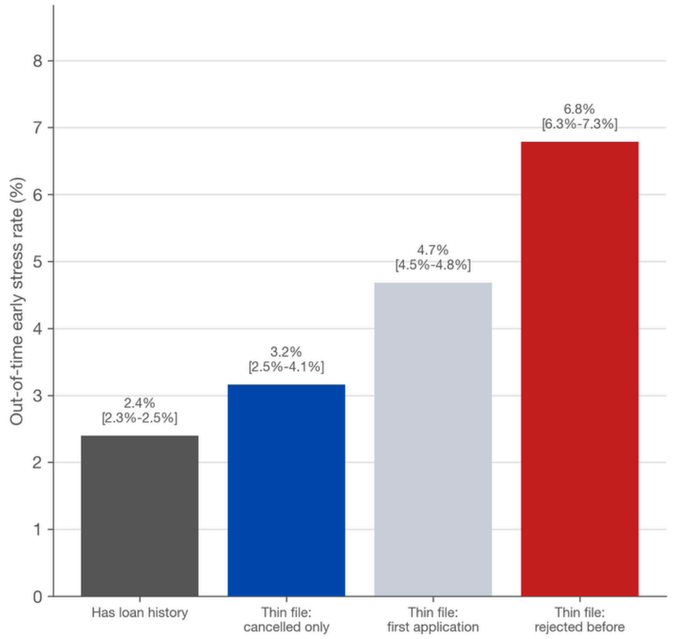


Figure 8: **Stress by history type: "no history" hides two opposite profiles**



Tier	Score	Customers	Share	Stress	vs avg
High	≥ 80	268,670	26.9%	1.13%	0.4×
Moderate	50 - 79	519,814	52.0%	3.10%	1.0×
Low	< 50	212,044	21.2%	6.25%	1.9×

Fairness check

- **Mean FHS**: women 65.1, men 66.5, it ranks within every group.
- Gender and age stay out of the score.

Insight: thin file ≠ stress

- **71.4%** of the Low tier has no signed loan.
- Rejected-before (156,891 median 46) get education and an affordability check, cancelled-only (25,529, median 62) get a service follow-up.

This answer tells us who is safe, but not who will come back. Challenge 2 do...

CHALLENGE 2 - CUSTOMER ENGAGEMENT SCORE

Most customers engaged only once around Tết 2025, and the strongest engagement comes from customers who contact us first.

***Figure 9*: Recency, frequency and product breadth of customers**

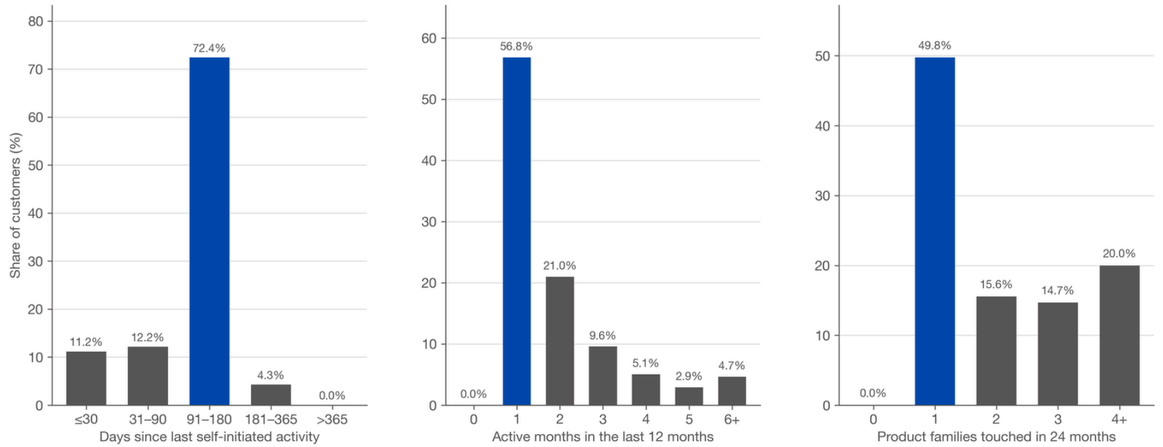
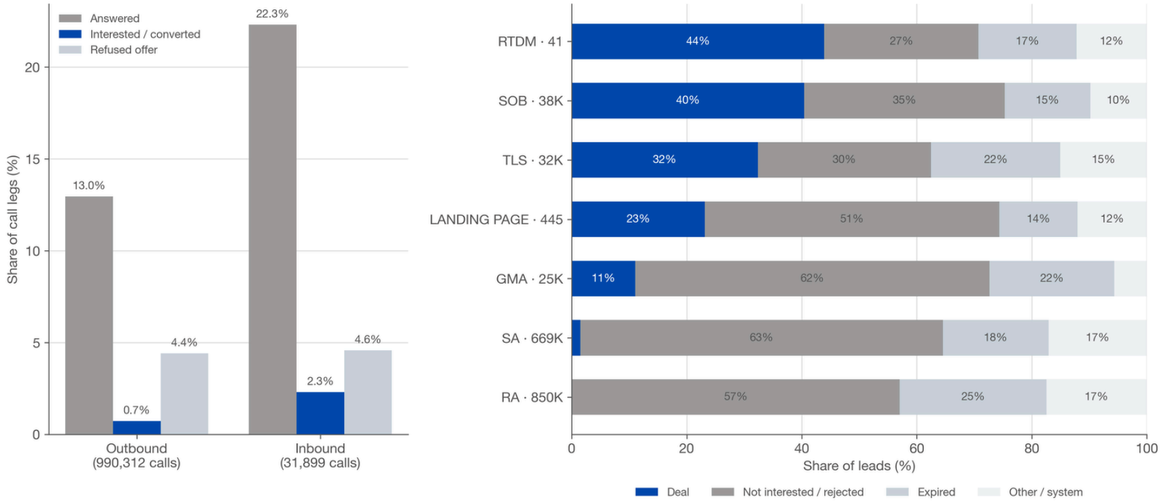


***Figure 10*: Call outcomes by direction and lead outcomes by source**



72.4%
last active in Tết window

Engagement is one off and narrow: Most customers have not acted since their application before Tết, 56.8% were active in a single month and 49.8% used one product family, which leaves a large base for low cost cross selling.

3.3x
better results when customers call us

Engagement is a conversation the customer starts: Inbound calls end interested or converted **2.3% vs 0.7%**; SOB and TLS close 32 to 40% of leads, while the 1.5M SA and RA leads are mostly refused (57 to 63%) or left to expire.

ACTION

For the score: Recency barely separates customers (AUC 0.50), so the score leans on **frequency** and **product breadth**, the two behaviours that still vary.

For the business: Prioritise channels where customers start the contact (SOB, inbound, app leads) and review who receives SA and RA offers, rather than calling more people.

CHALLENGE 2 - CUSTOMER ENGAGEMENT SCORE

Our five part engagement score predicts who will apply again, as High tier customers apply 2.9 times more often than Low tier customers.

Figure 11: Engagement score distribution and later outcomes by tier

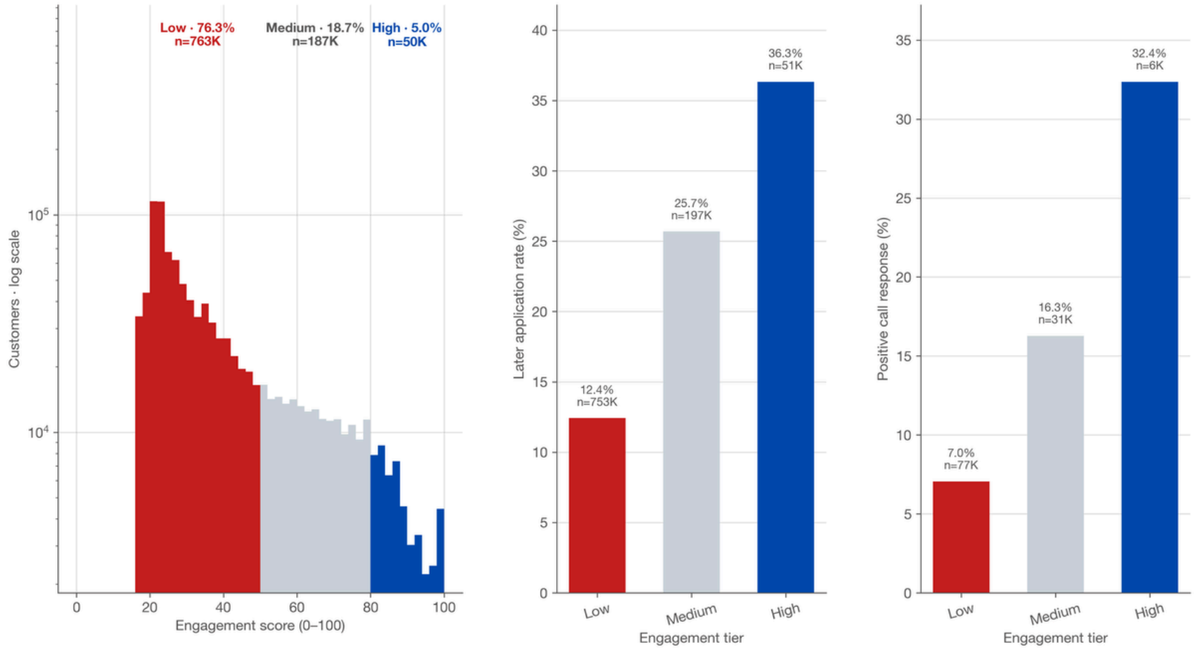


Table 6: How the Score is built

Component	Weight	What it measures	Evidence
E1 Digital	***20%***	App sessions / week or self-serve applications (12m)	AUC 0.57
E2 Recency	***25%***	Days since last activity, half-life 120 days	AUC 0.50 (flat in a Tết cohort)
E3 Frequency	***20%***	Active months in the last 12 months (6+ = full)	AUC 0.66
E4 Responsiveness	***20%***	Call pickup, positive reaction, lead replies, inbound calls	Lifts call conversion AUC 0.645 → 0.705
E5 Breadth	***15%***	Product families touched in 24 months (3+ = full)	AUC 0.63

Insight

Using a score frozen on 1 Feb 2025, High tier customers applied again by June 36.3% vs 12.4% for Low tier. Frequency and product breadth drive this ranking, and equal weights give almost the same result (AUC 0.653 vs 0.649), so it does not depend on our chosen weights.

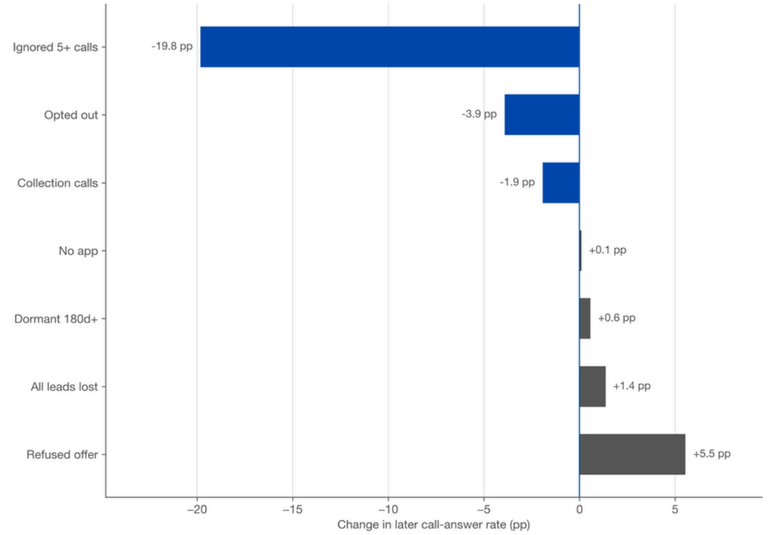
Implication

Send the **50K High tier** to service led offers, nurture the **187K Medium tier**, and keep the **763K Low tier** on low cost opt in channels.

CHALLENGE 2 - CUSTOMER ENGAGEMENT SCORE

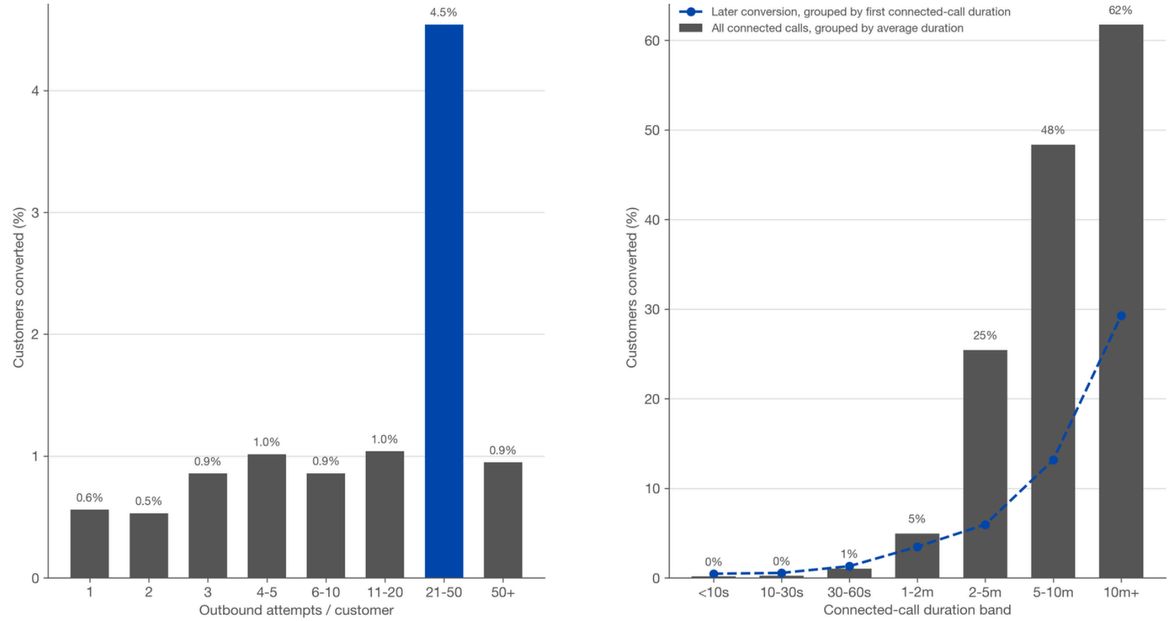
Engagement grows with the quality of each conversation rather than the number of calls, and ignored calls stay ignored.

Figure 12: Engagement score distribution and later outcomes by tier



Factor	Effect	Action
Customer calls in	3.3× better results	Prioritise inbound and SOB leads
Call over 2 minutes	25% to 62% conversion	Coach on conversation quality
Ignored 5+ calls	**−19.8 pp** answer rate	Stop calling, switch to SMS or app
Collection calls, opt outs	**−1.9 to −3.9 pp**	No sales calls
No app, dormancy	No real effect	Still reachable by phone

Figure 13: Conversion by number of calls and by call length



Insight & Implication

Insight

More calls barely help, since conversion stays at 0.5 to 1% however often we dial. Calls longer than two minutes, however, convert 25 to 62% of customers, while customers who ignored five calls answer only 3.3% of later calls versus 23.2% for others.

Implication

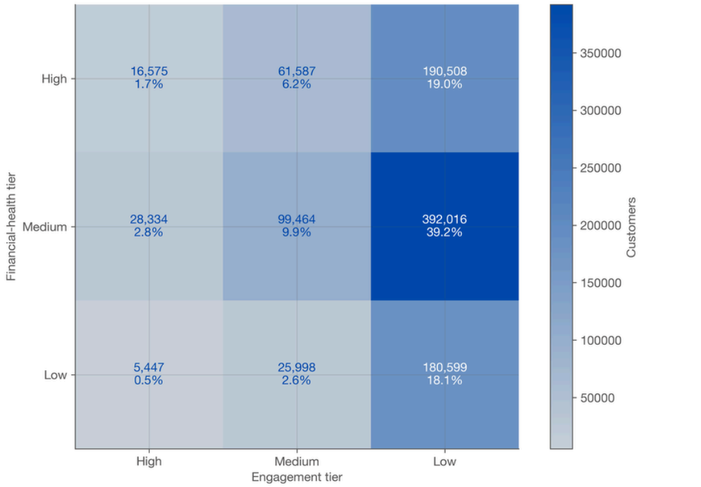
Measure teams on conversation quality, not call volume, and switch to SMS or app after five unanswered calls. Since engaged customers are not always financially safe, Challenge 3 adds financial health.

CHALLENGE 3 - CUSTOMER SEGMENTATION & NEXT BEST ACTION

Knowing who can repay is not enough; we also need to know who will come back, and only the two together tell us what to do.

Figure 14: Customers by health tier (rows) × engagement tier (columns)

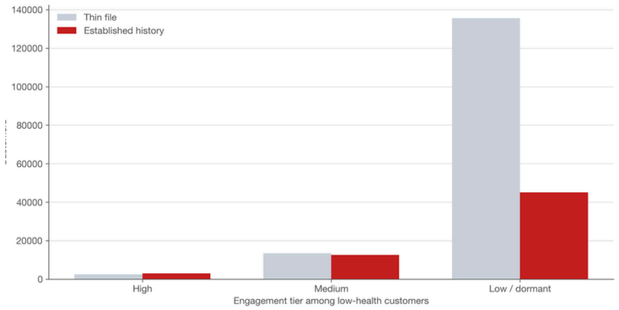
C1 · Can they repay? + **C2 · Will they come back?** → **C3 · Matrix**



Cell → next best action

High × High	needs-based review
High × Medium	product review + checks
High × Low	pre-Tết reminder or check-in
Moderate × High	tailored service review
Moderate × Medium	concise guidance
Moderate × Low	pre-Tết or utility reminder
Low × any	education or repayment support

Figure 15: Low-health customers by engagement: thin file (grey) vs established history (red)



151,447 thin files get education, not a stress label. **60,597** with real history get repayment support: **15,544** reachable now, **45,053** via proactive SMS/app.

1: GROW
205,960
20.6% of customers

Healthy and engaged: money made now
High engagement re-applies 36.3% vs 12.4% for Low; stress ≤ 3.1%.
→ **Action:** product and service reviews, after affordability checks.

2: RE-TIME
392,164
39.2% of customers

Healthy, one-time Tết borrowers: money waiting
87.6% were last active in Dec–Jan; they are seasonal, not lost.
→ **Action:** one reminder in early December, no spend now.

3: PROTECT
60,597
6.1% of customers

Low health with real loan history: money at risk
Low tier is stressed at 6.25%, 1.9× average.
→ **Action:** repayment support (15,544 reachable now, 45,053 by SMS/app), no new-credit offers.

HOLD: LOW COST
341,807
34.2% of customers

Thin files and other quiet customers
151,447 thin files get education; 190,360 quiet healthy customers get opt-in digital reminders. No call-centre time.

CHALLENGE 3 · CUSTOMER SEGMENTATION & NEXT BEST ACTION

Only one in five customers is ready to grow, while 39% should wait for the next Tết and 6% need repayment support first.

Table 7: Eleven actions in four groups (total 1,000,528 customers)

Group	Segment	Next best action & channel	Customers	Why it fits
GROW	High × High	Needs-based review after eligibility relationship	16,575	Best customers: keep them with service
GROW	High × Medium	Product review with suitability checks service	61,587	Healthy, responsive: the next sale
GROW	Moderate × High	Service review; affordability first service	28,334	Engaged, moderate health: serve, then lend
GROW	Moderate × Medium	Concise guidance, only on a clear need digital	99,464	Mixed signals: avoid call fatigue
RE-TIME	High/Mod. × Low, Tết	One opt-in reminder in early December digital	392,164	Seasonal borrowers: re-time, don't chase
PROTECT	Low · history × High	Repayment-support review; no promotion support	2,942	Engaged yet stressed: first queue to call
PROTECT	Low · history × Med	Support options + contact, no promotion support	12,602	Reachable, stressed: support first
PROTECT	Low · history × Low	Proactive SMS/app repayment reminders, no calls	45,053	Riskiest, only 9.5% ever answered
HOLD	Low · thin file	Neutral education, affordability first digital	151,447	No history is not a missed payment
HOLD	Moderate × Low	Opt-in account-utility reminder digital	120,674	Little activity: keep cost low
HOLD	High × Low	Opt-in digital check-in, quiet ≠ dissatisfied	69,686	Strong repayment evidence, low activity

Guardrail overrides on top of the matrix

33,506
SMS/app only: ignored 5+ sales calls

21,725
Overdue check before sales: collection calls ≠ distress (95% never late)

30,973
Thin file above Low: affordability check before any offer

355
Hard block: in arrears yet still offered a loan

Five decisions save 1.5 to 8.3bn VND a year and stop lending to customers who cannot repay.

#	DECISION & WHY	WORTH PER YEAR	HOW SURE	FIRST STEP
1	**Stop calling customers who never pick up** 37 of every 100 outbound calls go to 113,644 people who have never answered. After 5 tries, switch to SMS/app.	**1.2 - 4.9bn VND** saved, net of SMS cost	**Medium / test first**	Pilot the 5-call cap against a control group
2	**Spend lead budget on self-initiated customers** RA makes 54% of all leads but under 1% of deals: 2,900 leads per deal vs 2.4 when customers ask themselves.	**0.35 - 3.45bn VND** saved; ≤ 238 deals given up	**Medium / weak credit**	Filter RA lists using the health and engagement scores
3	**Be stricter on first-time appliance loans and 0%** Expected losses eat 17–63% of what first-time appliance (CD) customers pay; first motorbike loans need more down payment.	**≈ 7.5bn VND** expected loss at stake	**Medium / fees unseen**	Keep 0% for customers who already repaid well
4	**Chase late payments differently per product** A first missed payment ends in write-off for 68% of cash-at-counter loans but only 8% of credit cards.	**Fewer write-offs** not priced	**High / direct data**	Reminders for cards and BNPL; early call for cash loans
5	**Never offer a new loan to someone already behind** 355 customers in arrears still received loan offers within 90 days.	**No lending into arrears** not priced	**High / direct data**	Hard block in the offer system; fix call-date logging

Bottom line: decisions 1 and 2 alone save 1.5 - 8.3bn VND a year. Each starts as a pilot with a control group and scales only if it works.

About 2.8bn VND a year leaks through unanswered calls, low-yield RA leads and underpriced first-time appliance loans → The money is here

DECISION 1 · CALLS

37 of every 100 calls reach no one



37.4% never answered 62.6% other

113,644 customers received 5 or more calls and never picked up, which adds up to **1,449,087 of 3,875,171** outbound calls. Stopping after the 5th call would free about **528,000 dials a year** for engaged customers.

Net saving per year

1.18bn	**2.76bn**	4.88bn
low	base	high

DECISION 2 · LEADS

Most leads come from the source that closes least

RA · pushed leads

53.9% of leads

0.8% of deals

SOB · customer asks first

2.6% of leads

49.1% of deals

Cost of one recorded deal: about **5.8m VND** from RA vs **4.8k VND** from SOB (base case).

Caution: ~80% of customers sign something within 90 days anyway, so this is recorded performance, not proven impact.

DECISION 3 · FIRST-TIME CD

Appliance loans pay too little to absorb risk

First-time CD: 17–63% VND



First-time motorbike: 8–16% VND



Repeat cash loan: 2–3 % VND



Repeat CD only breaks even before retailer fees (0.07% margin, base).
Keeping 0% for proven payers is safe: CD with < 10% down runs at 1.46% stress.

Assumptions (low / base / high) Cost per call 3k / 6k / 10k VND · SMS 300 VND · cost per lead 0.5k / 2k / 5k VND · loss when a loan defaults 45% / 60% / 75% · funding 6% / 8% / 10%. A relative comparison, not a full P&L: no opex, collection cost or partner fees.